23/7/26

Exp 1: Study of various Networks Commands used in Linux And Windows.

- arp -a

ARP is a short form of address resolution protocol. It shows ip address of your computer and MAC address of your routes

- Op: - gateway (10.0.2.2) at 52.54:00:12:35:02 (ether or uposs)
- hostname: displays the name of your compiler
- Op: localhost localdomain
- ipconfig/all: displays detailed networks configuration for all network adapts on your computer

Window IP Configuration

Host name: Sweta-s-k

Primary Dns System....:

Node Type

Ip routing enabled....: No

WINS proxy enables..: No

nbstat -a: helps to solve problem with Net BIOS resolution

OP: Displays protocol statistics and current TCP/IP connection using NBT

-a (adapter status)

-A (adapter status).

Pathping: The path it take to reach destination and where networks fails in between.

op: -g host-list loose source route along host-list
-h maximum hops maximum number of hops to search for target

ping: Packet Internet, helps to check connectivity between two points

op: -t ping the specified host until stopped
To see statistics and continue -type control. Break:

route: route command is used to show/manipulate Ip routing table

op: -f clears the routing table of all gateway entries. If this is used in conjuction with one of commands the tables are cleared prior to running the command.

Linux Network commands.

ip: It is used to troubleshooting existing systems the ip command can show address information manipulative routing, plus display network various devices, interfaces

ip <OPTION><OBJECT> <COMMAND>

@ to show ip address assigned to an interface on your server

ⓑ ip address add 192.168.1.254/24 dev enp0s3

ⓒ ip address del 192.168.1.245/24 dev enp0s3

2) ifconfig: It was a staple in many system admin tool belt for configuration and trouble shooting between networks

op: enp0s3 flags 0183 <op, BROADCAST, RUNNING, MULTICASTING> mtu 1500
inet 10.0.2.15 netmaste
2555 2555 255.0...

3) mtr: MTR is a program with a command line interface that server as a network diagnostic and troubleshooting tool. This command combines the functionality of the ping and traceroute command. mtr.

op: my traceroute. [v0.94]

	Host	Packets		Pings		
		Loss%	Snt	Last	Avg	Best	Wrst
1	Local Host	0.0%	122	0.1	0.2	0.1	1.9

@ # mtr google.com

op: Host
1. gateway
2. (waiting for reply).

(b) # mtr -g google.com

op: -F filename FILE

-u -- tcp

4) tcpdump: The tcpdump command is designed to capturing and displaying packet.

op:

tcpdump: enpass: you don't have permission to capture on that device.

5) Ping:

ping is a tool that verifies IP-level connectivity to another TCP/IP computer by sending Internet control message

→ Command ping google.com

op: PING google.com (142.200.18.2.78) 56(84) bytes of data 64 byte from 1emaa-ax-in.t4 1e100.net (142.250.182.78): dump-seq =1 ttl = 112 time = 9.38 ms.

Result:

Thus the network commands using linux and windows are success fully executed and the output is verified.